

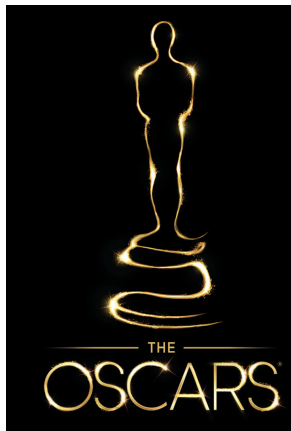
42.510 Statistical Learning for Data Science

Discrete choice models

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Week 4, Lecture 2

The Academy Awards



- Popularly known as the Oscars.
- Started in 1929.
- The Academy of Motion Picture Arts and Sciences (AMPAS).
- Honors excellence in acting, directing, writing, editing, etc.
- Primarily English language movies
- ... made by a US production house ...

Award: nomination and voting

- Eligible voters: More than 10,000 members of AMPAS from different disciplines (actors, writers, designers, directors) as of 2025.
- Each branch of AMPAS vote to determine nominees from their own category.
 - In special cases like Best Picture, everyone votes.
- Typically the top 5 voted in each category are chosen to be Oscar nominees.
- Nominees announced in around January.
- All members (in most cases) vote for deciding final winners.
- Oscars ceremony occurs late February , early March.

Effect of the Academy Awards

- Viewership (has gone down over the years)
 - ▶ 1998 (70th Academy Awards): 57 million viewers
 - ★ Best picture: Titanic
 - ▶ 2015 (87th Academy Awards): 37 million viewers
 - ★ Best picture: Birdman
 - ▶ 2020 (92nd Academy Awards): 23.6 million viewers
 - ★ Best picture: Parasite
 - ▶ 2021 (93rd Academy Awards): 10.4 million viewers
 - ★ Best picture: Nomadland
 - ▶ 2025 (97th Academy Awards): 19.5 million viewers
 - ★ Best picture: Oppenheimer
- Significant impact on ticket sales: [Boxofficemojo](#)
- Winning an Oscar leads to many better deals for the parties involved.
- An early prediction may lead to having an arbitrage opportunity for some.

How can one predict the Oscars

- Is it possible to predict the Oscars awards with any degree of accuracy?
- Many predictions are there using the buzz of awards, and critics choices.
- Develop a data driven approach.
- Follow: *Applying Discrete Choice models to predict Academy Award winners* by I. Pardoe and D.K. Simonton in *JRSS(B)*, 2008.
- Prediction is done after the 5 nominees are announced.
- Prediction for four categories:
 - ▶ Best picture, Best director, Best actor and Best actress.
- The data will have:
 - ▶ **years** $i \in \{1951, \dots, 2007\}$.
 - ▶ **options** $k \in \{1, \dots, 5\}$.

Binary choice model

- n = Number of observations
- p = Number of predictor variables
- y = **Observed** response variable, taking values 0 or 1 (No or Yes).
- $x_1, \dots, x_p$ = attributes of choice, or, explanatory variables.
- We model the responses as

$$y^* = \beta_0 + \sum_{j=1}^p \beta_j x_j + \epsilon =: \boldsymbol{\beta}^\top \mathbf{x} + \epsilon.$$

- y^* is an unobserved latent variable, ϵ is the error term and $\boldsymbol{\beta}$ are parameters.
- The relationship between y and y^* is:

$$y = \begin{cases} 1 & \text{if } y^* \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Binary choice model

- Y : random variable representing the responses.
- Let $\epsilon \sim F(\cdot)$ and be symmetric.

$$\begin{aligned}\mathbb{P}(Y = 1|\mathbf{x}) &= \mathbb{P}(Y^* \geq 0|\mathbf{x}) = \mathbb{P}(\beta^\top \mathbf{x} + \epsilon \geq 0|\mathbf{x}) \\ &= \mathbb{P}(\epsilon \geq -\beta^\top \mathbf{x}|\mathbf{x}) = \mathbb{P}(\epsilon \leq \beta^\top \mathbf{x}|\mathbf{x}) = F(\beta^\top \mathbf{x}).\end{aligned}$$

- With a choice of $F(z) = \frac{e^z}{1+e^z}$, $z \in \mathbb{R}$ we get,

$$\mathbb{P}(Y = 1|\mathbf{x}) = \frac{e^{\beta^\top \mathbf{x}}}{1 + e^{\beta^\top \mathbf{x}}}.$$

- This is the logistic regression model we have seen previously.

Multinomial choice model

- $k \in \{1, \dots, K\}$ choices (alternatives)
- $i \in \{1, \dots, n\}$ consumers (observations)

- The utility of consumer i for alternative k is,

$$U_{ik} = \beta^\top \mathbf{x}_{ik} + \epsilon_{ik}.$$

- β : weight given to the observable information $\mathbf{x}_{ik}$.
- $\mathbf{x}_{ik}$: aspects specific to individual i (e.g., demographic information) and alternative k (e.g., price of a product).
- ϵ_{ik} : noise term.
- If Y_i denotes the choice of individual i then

$$\Pr(Y_i = k) = \Pr(U_{ik} > U_{il}, \forall l \neq k).$$

Multinomial choice

- Let us write:

$$U_{ik} = \beta^\top \mathbf{x}_{ik} + \epsilon_{ik} = V_{ik} + \epsilon_{ik}$$

where V_{ik} 's are fixed and ϵ_{ik} 's are random.

- Assume: ϵ_{ik} 's are iid Gumbel distributed. Hence

$$\text{CDF : } F(y) = e^{-e^{-y}}, \quad y \in \mathbb{R},$$

$$\text{PDF : } f(y) = e^{-y} e^{-e^{-y}}, \quad y \in \mathbb{R}.$$

- It was shown by McFadden (1974) that,

$$\mathbb{P}(Y_i = k) = \frac{e^{V_{ik}}}{\sum_{\ell=1}^K e^{V_{i\ell}}}.$$

- Called: **Conditional logit model**, also referred to as *multinomial logit model*.

Multinomial choice: other variants

- **Multinomial logistic regression**

- Assume only individual specific data $\mathbf{x}_i$ and suppose we want to categorize the individual into one of several categories:

$$\Pr(Y_i = k | \mathbf{x}_i) = \frac{e^{\beta_k^\top \mathbf{x}_i}}{\sum_{l=1}^K e^{\beta_l^\top \mathbf{x}_i}}.$$

- ▶ Here β_k is a vector of weights corresponding to category k .
- In some cases, we may have **alternate specific constants** ASC_k for each alternative.

$$U_{ik} = ASC_k + \beta^\top \mathbf{x}_{ik} + \epsilon_{ik}.$$

Multinomial choice model: estimation

- Observations:

- ▶ $\mathbf{x}_{ik}$ = attributes of individual i and alternative k .
- ▶ $z_{ik} = \begin{cases} 1 & \text{if individual } i \text{ chooses } k \\ 0 & \text{otherwise. (Note } y_i = k \iff z_{ik} = 1) \end{cases}$

- Optimal β obtained by solving:

$$\max_{\beta} \underbrace{L(\beta)}_{\text{Likelihood given } \beta} = \max_{\beta} \prod_{i=1}^n \prod_{k=1}^K \mathbb{P}(Y_i = k)^{z_{ik}}$$

- Taking logarithms, we can be reformulate as maximizing the log-likelihood.

$$\max_{\beta} \underbrace{LL(\beta)} = \max_{\beta} \sum_{i=1}^n \sum_{k=1}^K z_{ik} \log \left(\frac{e^{\beta^\top \mathbf{x}_{ik}}}{\sum_{l=1}^K e^{\beta^\top \mathbf{x}_{il}}} \right)$$

- This problem can be efficiently solved as the objective function is concave.

- **Akaike's Information Criterion (AIC)**

$$AIC = -2LL(\hat{\beta}) + 2p,$$

where LL: Log-likelihood, p : no. of parameters, $\hat{\beta}$: estimated value of the parameters. Smaller the AIC, the better.

- **Likelihood ratio index (McFadden's index)**

$$\rho = 1 - \frac{LL(\hat{\beta})}{LL(\mathbf{0})},$$

where $LL(\mathbf{0})$ is the log-likelihood when all the parameters are set to 0 (no feature is used as a predictor).

- **Percent correctly predicted**

- ▶ This identifies the alternative with the highest probability for each individual observation and determining whether or not this was what the actual choice was.

References

- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 4.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 4.
- Pardoe et al. (2008), *Applying discrete choice models to predict Academy Award winners*.